Understanding Contributions to Reductions in Cost of Transport with Stiffness-Customized Ankle-foot Orthosis: A Case Series

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INTRODUCTION

Individuals post-stroke often experience plantar flexor weakness, limiting their ability to control stance-phase shank forward rotation in gait. Passive-dynamic ankle-foot orthoses (PD-AFOs) are commonly prescribed to individuals post-stroke as PD-AFO bending stiffness has been shown to assist weakened plantar flexors (Koller, 2021a). Our recent research showed stiffness-customized PD-AFOs enable individuals post-stroke to reduce their lower extremity mechanical cost of transport (COT) when walking compared to both walking without any AFO and with their standard-of-care (SOC) AFO (Koller, 2021b). The purpose of this case-series was to begin to investigate how individuals achieve this COT reduction. The Constituent Lower Extremity Work (CLEW) approach (Ebrahimi, 2017) was used to quantify and visualize how energetics of the lower extremity constituents (distal foot, ankle, knee, hip) changed with PD-AFO use. We hypothesized COT reductions would be driven by paretic ankle work decreases.

METHOD

Subjects: Two individuals six+ months post-stroke participated. Participant 1 (P1): Female, 65 yrs, 1.65 m, 89.8 kg, 0.42 m/s self-selected walking speed. P2: Female, 75 yrs, 1.58 m, 67.4 kg, 0.36 m/s.

Apparatus: Instrumented gait analysis.

Procedures: An instrumented gait analysis was conducted as participants walked without any AFO and then with their SOC AFO at their self-selected waking speed determined via a 10 Meter Walk Test without an AFO. A PD-AFO was stiffness customized and manufactured for each individual based on the No AFO data. Participants were given the PD-AFO to wear for four weeks, then they returned to the lab for an instrumented gait analysis with the PD-AFO at the same self-selected walking speed from the initial visit.

Data Analysis: Data were processed and analyzed in Visual 3D. COT was calculated per limb for each condition as positive limb work (summed hip, knee, ankle, and distal foot, all normalized by body mass) summed with the absolute value of negative limb work over the gait cycle, scaled by stride length. Finally, CLEW analysis was performed (Ebrahimi 2017).

RESULTS

P1 used a hinged SOC AFO with plantarflexion stop and P2 used a solid SOC AFO. Both participants reduced their paretic and nonparetic limb COT with PD-AFO use compared to the No AFO and SOC AFO (except P2's No AFO nonparetic work data were not collected due to technical difficulty). P1 had a reduction in nonparetic limb COT but no change in paretic limb COT with the SOC AFO compared to No AFO. P2 had an increase in paretic limb COT with the SOC AFO compared to No AFO. Some of P1's work constituents decreased while others increased with PD-AFO use, although the net decrease in paretic limb COT with PD-AFO use was driven by a reduction in positive paretic ankle work (↓44%) and negative paretic hip work (↓48%) during stance. P1's nonparetic limb and SOC AFO condition showed similar patterns. P2 showed substantial (16-70%) reductions in all constituent work values with PD-AFO use compared to No AFO, with positive paretic ankle work (↓71%) and negative paretic knee work (↓66%) during stance showing the greatest reductions (Figure 1). P2's increase in paretic limb COT with SOC AFO use was driven by an increase in both positive (↑21%) and negative (↑42%) stance-phase ankle work.

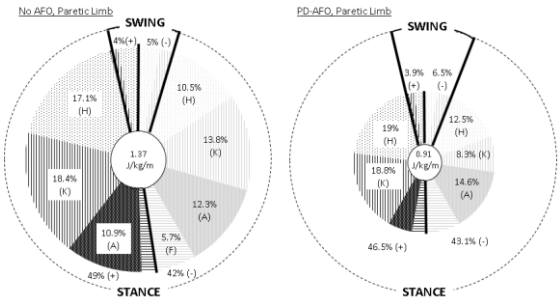

Figure 1. P2's No AFO (left) and PD-AFO (right) CLEW charts. Note: percentages in chart indicate % each constituent contributed to limb COT, which differs from percent changes from No AFO reported in results above.

DISCUSSION

While there was variability between the participants, the ankle appeared to be a prime driver of changes in COT, although the hip/knee also played important roles. Future work will analyze additional participants.

CONCLUSION

These PD-AFOs enable COT reductions through decreases in paretic ankle work, combined with other, smaller constituent changes.

CLINICAL APPLICATIONS

Understanding how lower extremity energetics during gait change with stiffness-customized PD-AFO, and SOC AFO use, provides valuable insight into how the post-stroke musculoskeletal system interacts with AFOs and will advance efforts to optimize the design and prescription of AFOs.

REFERENCES

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